Scott's Radiation Activation Law

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Abstract

This paper investigates a nonlinear differential equation with a factorially normalized, Erlang-type forcing term:

$$\frac{dN}{dt} = G_0 \frac{\lambda^m t^{m-1} e^{-\lambda t}}{(m-1)!} - k_1 N - \beta N^2.$$

Here, $N(t)$ represents a time-dependent state variable, the first term describes a transient external input, and the remaining terms represent linear and quadratic losses. We establish existence, uniqueness, nonnegativity, and global boundedness of solutions under natural assumptions on the parameters and initial condition. We derive an integral representation, obtain an explicit solution in the absence of quadratic loss, and establish bounds for the accumulated input and the resulting state variable. We also examine the long-time behavior of the model and derive a finite-time approximation framework. The results provide a mathematical foundation for studying systems subject to a transient input and nonlinear dissipation.

Keywords: nonlinear differential equation; Erlang distribution; factorial normalization; existence and uniqueness; bounded solutions; nonlinear dissipation.

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1 Introduction

Differential equations provide a general framework for describing systems whose states change in response to external inputs and internal loss mechanisms. In many applications, an external input is concentrated over a finite period rather than remaining constant indefinitely. The subsequent evolution may depend on both linear and nonlinear dissipation.

This paper considers the initial-value problem

$$\frac{dN}{dt} = G_0 \frac{\lambda^m t^{m-1} e^{-\lambda t}}{(m-1)!} - k_1 N - \beta N^2, \qquad t \geq 0, \tag{1}$$

subject to

$$N(0) = N_0. \tag{2}$$

The forcing term has the form of a scaled Erlang probability density. The parameters determine the amplitude and timing of the input, while the terms proportional to N and N^2 describe linear and quadratic losses, respectively.

The main questions are whether the initial-value problem has a unique solution, whether nonnegative initial data remain nonnegative, whether solutions can grow without bound, and how the model behaves over long time intervals. We also investigate the special case in which the quadratic loss is absent and establish estimates useful for numerical approximation.

The analysis uses standard methods from ordinary differential equations, including local existence and uniqueness, comparison arguments, and integral representations; see Coddington and Levinson [1] and Teschl [2].

2 Definition and Basic Properties

2.1 The mathematical model

Definition 2.1 (Factorial-Erlang Forcing Model)**.** Let m be a positive integer and let $G_0 \geq 0$, $\lambda > 0$, $k_1 \geq 0$, $\beta \geq 0$, and $N_0 \geq 0$. The model is the initial-value problem

$$\begin{cases} \dfrac{dN}{dt} = G_0 \dfrac{\lambda^m t^{m-1} e^{-\lambda t}}{(m-1)!} - k_1 N - \beta N^2, & t \geq 0, \\ N(0) = N_0. \end{cases}$$

The variables and parameters have the following roles:

- $N(t)$ is the state variable.
- N_0 is its initial value.
- G_0 controls the total magnitude of the input.
- λ is the rate parameter of the input.
- m is a positive integer controlling the shape of the forcing profile.
- k_1 is the linear loss coefficient.
- β is the quadratic loss coefficient.

The nonnegative parameter assumptions are appropriate when N represents a population, concentration, or other nonnegative quantity subject to losses. Different applications may require different parameter assumptions.

For convenience, define the forcing function

$$F_m(t) = G_0 \frac{\lambda^m t^{m-1} e^{-\lambda t}}{(m-1)!}. \tag{3}$$

Equation (1) can then be written as

$$N'(t) = F_m(t) - k_1 N(t) - \beta N(t)^2. \tag{4}$$

2.2 Properties of the forcing term

The forcing function is nonnegative for $t \geq 0$. For $G_0 > 0$, it is positive for $t > 0$. Its total integral is

$$\int_0^\infty F_m(t)\, dt = G_0. \tag{5}$$

Indeed, the substitution $u = \lambda t$ gives

$$\int_0^\infty F_m(t)\, dt = \frac{G_0}{(m-1)!} \int_0^\infty u^{m-1} e^{-u}\, du.$$

Using the factorial identity

$$\int_0^\infty u^{m-1} e^{-u}\, du = (m-1)!,$$

we obtain Equation (5).

More generally, the cumulative input delivered by time t is

$$A_m(t) = \int_0^t F_m(s)\,ds = G_0\left[1 - e^{-\lambda t}\sum_{j=0}^{m-1}\frac{(\lambda t)^j}{j!}\right]. \tag{6}$$

This identity follows from the standard integral formula for the Erlang density. In particular,

$$0 \le A_m(t) \le G_0, \qquad \lim_{t\to\infty} A_m(t) = G_0.$$

Thus, the forcing term supplies a finite total input, even though the model is defined over the entire nonnegative time axis.

2.3 Interpretation of the loss terms

The linear loss term is $-k_1 N$. When considered alone, it produces exponential decay at rate k_1.

The quadratic loss term is $-\beta N^2$. For $\beta > 0$, this term increases in magnitude quadratically with the state variable and can represent density-dependent loss or other nonlinear dissipation.

When both coefficients vanish, the model reduces to $N' = F_m(t)$. When the quadratic coefficient vanishes but the linear coefficient remains, the equation becomes linear and can be solved explicitly. The general case with both loss terms present is nonlinear.

3 Existence and Uniqueness

3.1 Local existence and uniqueness

Theorem 3.1. *For the parameter values specified in the definition, the initial-value problem (1)–(2) has a unique local continuously differentiable solution.*

Proof. Define

$$f(t, N) = F_m(t) - k_1 N - \beta N^2.$$

Since m is a positive integer and $\lambda > 0$, the function F_m is continuous on $[0, \infty)$. Consequently,

f is continuous in both variables.

Furthermore,

$$\frac{\partial f}{\partial N}(t, N) = -k_1 - 2\beta N$$

is continuous for all $t \geq 0$ and $N \in \mathbb{R}$. Hence f is locally Lipschitz continuous in N.

The standard existence and uniqueness theorem for ordinary differential equations therefore gives a unique local solution through the initial point $(0, N_0)$ [1, 2]. □

Local existence does not, by itself, guarantee that a solution exists for every $t \geq 0$. We establish that stronger conclusion by proving nonnegativity and boundedness.

3.2 Nonnegativity of solutions

Theorem 3.2. *Suppose $G_0 \geq 0$, $\lambda > 0$, $k_1 \geq 0$, $\beta \geq 0$, and $N_0 \geq 0$. Every solution of the initial-value problem remains nonnegative throughout its interval of existence.*

Proof. The equation can be written in the form

$$N'(t) + \Big(k_1 + \beta N(t)\Big)N(t) = F_m(t).$$

Multiplying by the integrating factor associated with the coefficient $k_1 + \beta N(t)$ gives

$$\begin{aligned} N(t) = N_0 \exp\left(-\int_0^t [k_1 + \beta N(u)]\, du\right) \\ + \int_0^t F_m(s) \exp\left(-\int_s^t [k_1 + \beta N(u)]\, du\right) ds. \end{aligned} \tag{7}$$

This identity holds along any existing solution. The exponential factors are positive, and $F_m(s) \geq 0$. Since $N_0 \geq 0$, both terms on the right-hand side are nonnegative. Therefore $N(t) \geq 0$. □

The representation in Equation (7) is an identity along the solution, rather than an explicit closed-form solution, because the exponent still depends on N.

3.3 Global existence and boundedness

Theorem 3.3. *Under the same parameter assumptions, the unique solution exists for every $t \geq 0$ and satisfies*

$$0 \leq N(t) \leq N_0 + A_m(t) \leq N_0 + G_0. \tag{8}$$

Proof. Nonnegativity has already been established. Therefore, the loss terms satisfy

$$-k_1 N(t) - \beta N(t)^2 \leq 0.$$

It follows that

$$N'(t) \leq F_m(t).$$

Integrating from 0 to t gives

$$N(t) - N_0 \leq \int_0^t F_m(s)\, ds = A_m(t).$$

Equation (6) yields

$$N(t) \leq N_0 + A_m(t) \leq N_0 + G_0.$$

Thus the solution remains in the bounded interval $[0, N_0 + G_0]$. The right-hand side of the differential equation is continuous in time and locally Lipschitz continuous in the state variable. Since the solution cannot become unbounded in finite time, the continuation theorem for ordinary differential equations implies that it extends to every $t \geq 0$. □

This theorem establishes global existence without requiring an explicit formula for the nonlinear solution.

4 Integral Representation and Special Cases

4.1 An integral identity

Integrating Equation (4) from 0 to t gives the identity

$$\begin{aligned} N(t) = N_0 + A_m(t) - k_1 \int_0^t N(s)\, ds \\ - \beta \int_0^t N(s)^2\, ds. \end{aligned} \tag{9}$$

This representation separates the accumulated input from the cumulative linear and quadratic losses. It also provides a useful starting point for numerical methods and analytical estimates.

The identity is exact, but it is not an explicit solution in the general nonlinear case because the integrals depend on the unknown state N.

4.2 The case of no quadratic loss

When $\beta = 0$, Equation (1) becomes

$$N'(t) + k_1 N(t) = F_m(t).$$

Multiplication by $e^{k_1 t}$ and integration yield the explicit solution

$$N(t) = N_0 e^{-k_1 t} + \int_0^t e^{-k_1(t-s)} F_m(s)\, ds. \tag{10}$$

Substituting the forcing function gives

$$\begin{aligned} N(t) = {} & N_0 e^{-k_1 t} \\ & + \frac{G_0 \lambda^m e^{-k_1 t}}{(m-1)!} \int_0^t s^{m-1} e^{-(\lambda - k_1)s}\, ds. \end{aligned} \tag{11}$$

This formula remains valid when $k_1 = \lambda$. In that case, the integral simplifies to

$$\int_0^t s^{m-1}\, ds = \frac{t^m}{m},$$

so that

$$N(t) = N_0 e^{-\lambda t} + \frac{G_0 \lambda^m t^m e^{-\lambda t}}{m!}. \tag{12}$$

When $k_1 \neq \lambda$, the integral in Equation (11) can also be expressed using the incomplete gamma function.

4.3 The case of no losses

If $k_1 = \beta = 0$, then

$$N'(t) = F_m(t).$$

Using the initial condition and integrating directly, we obtain

$$N(t) = N_0 + G_0 \left[1 - e^{-\lambda t} \sum_{j=0}^{m-1} \frac{(\lambda t)^j}{j!} \right]. \tag{13}$$

In particular,

$$\lim_{t \to \infty} N(t) = N_0 + G_0.$$

This limiting value represents the initial state plus the total accumulated input when no loss mechanism is present.

4.4 The general nonlinear case

For $\beta > 0$, the quadratic term makes the equation nonlinear. The substitution

$$N(t) = \frac{u'(t)}{\beta u(t)}$$

transforms the equation, wherever the substitution is defined, into

$$u''(t) + k_1 u'(t) - \beta F_m(t) u(t) = 0. \tag{14}$$

To verify this transformation, differentiation gives

$$N'(t) = \frac{u''(t)}{\beta u(t)} - \beta N(t)^2.$$

Substitution into Equation (1) cancels the quadratic terms and yields Equation (14).

For $\beta > 0$, the initial conditions corresponding to $N(0) = N_0$ can be chosen as

$$u(0) = 1, \qquad u'(0) = \beta N_0.$$

The resulting solution must be considered on intervals where $u(t) \neq 0$.

This transformation provides an alternative linear second-order formulation, although it does not generally produce an elementary closed-form expression for $N(t)$.

5 Long-Time Behavior

5.1 Decay of the forcing term

For every fixed positive integer m,

$$\lim_{t\to\infty} F_m(t) = 0.$$

Indeed, exponential decay dominates the polynomial factor t^{m-1}.

Moreover, the total input is finite:

$$\int_0^\infty F_m(t)\,dt = G_0.$$

Thus, the external forcing becomes small at late times, and its remaining cumulative contribution tends to zero.

5.2 Decay when linear loss is present

Theorem 5.1. *If $k_1 > 0$, then every solution satisfying the parameter assumptions in Section 2 obeys*

$$\lim_{t\to\infty} N(t) = 0.$$

Proof. Because $N(t) \geq 0$ and $\beta \geq 0$,

$$N'(t) + k_1 N(t) = F_m(t) - \beta N(t)^2 \leq F_m(t).$$

Multiplying by $e^{k_1 t}$ and integrating yields

$$N(t) \leq N_0 e^{-k_1 t} + \int_0^t e^{-k_1(t-s)} F_m(s)\,ds.$$

The first term tends to zero because $k_1 > 0$. For the integral, fix $T > 0$ and split it as

$$\int_0^t e^{-k_1(t-s)} F_m(s)\,ds = \int_0^T e^{-k_1(t-s)} F_m(s)\,ds + \int_T^t e^{-k_1(t-s)} F_m(s)\,ds$$

for $t \geq T$.

The first integral tends to zero as $t \to \infty$, since its interval is fixed. The second is bounded by

$$\int_T^\infty F_m(s)\, ds,$$

which can be made arbitrarily small by taking T sufficiently large. Therefore the entire integral tends to zero, proving the result. □

5.3 Decay when only quadratic loss is present

Theorem 5.2. *Suppose $k_1 = 0$, $\beta > 0$, and the other parameter assumptions remain in force. Then*

$$\lim_{t\to\infty} N(t) = 0.$$

Proof. The equation becomes

$$N'(t) = F_m(t) - \beta N(t)^2.$$

Since $N(t) \geq 0$, integration gives

$$\beta \int_0^t N(s)^2\, ds = N_0 + A_m(t) - N(t) \leq N_0 + G_0.$$

Consequently,

$$\int_0^\infty N(s)^2\, ds \leq \frac{N_0 + G_0}{\beta} < \infty.$$

Also,

$$|N'(t)| \leq F_m(t) + \beta(N_0 + G_0)^2,$$

so N' is bounded. Hence N is uniformly continuous on $[0, \infty)$.

A nonnegative, uniformly continuous function whose square is integrable on $[0, \infty)$ must tend to zero. Otherwise, there would be a positive number ε and a sequence of arbitrarily late times at which $N(t) \geq \varepsilon$. Uniform continuity would then force intervals of uniformly positive length on which $N \geq \varepsilon/2$, contradicting square integrability. Therefore $N(t) \to 0$. □

Together, these results show that either form of nonnegative dissipation is sufficient to drive the state toward zero after the transient input has passed. If both loss coefficients vanish, the limiting value instead equals $N_0 + G_0$, as shown in Equation (13).

6 Numerical Approximation

6.1 A finite-time integral estimate

Equation (9) gives

$$N(t) = N_0 + A_m(t) - k_1 \int_0^t N(s)\, ds - \beta \int_0^t N(s)^2\, ds.$$

Since all terms are nonnegative except for the explicit subtraction signs, it follows that

$$0 \le N(t) \le N_0 + A_m(t).$$

This bound can be used to check numerical calculations. A computed trajectory that becomes negative or exceeds $N_0 + A_m(t)$ under the stated parameter assumptions is inconsistent with the analytical results.

6.2 An Euler approximation

For a step size $h > 0$, define $t_j = jh$ and approximate the solution using the explicit Euler method:

$$N_{j+1} = N_j + h\left[G_0 \frac{\lambda^m t_j^{m-1} e^{-\lambda t_j}}{(m-1)!} - k_1 N_j - \beta N_j^2\right]. \tag{15}$$

The initial approximation is N_0. For sufficiently small step sizes on a fixed finite interval, the explicit Euler method converges to the exact solution under the standard regularity assumptions for this differential equation.

However, the explicit Euler method does not automatically preserve nonnegativity for arbitrary step sizes. Numerical implementations should therefore use an appropriate step size or a numerical method designed to preserve the relevant qualitative properties.

6.3 Local truncation error

Let

$$f(t, N) = F_m(t) - k_1 N - \beta N^2.$$

For a sufficiently smooth exact solution, Taylor's theorem gives

$$N(t + h) = N(t) + hN'(t) + \frac{h^2}{2} N''(\xi)$$

for some $\xi \in (t, t + h)$.

Since $N'(t) = f(t, N(t))$, the Euler update differs from the exact solution by a local truncation error of order $O(h^2)$, provided the relevant second derivative is bounded on the interval under consideration. The resulting global error on a fixed finite interval is of order $O(h)$.

These statements concern finite-time numerical approximation; they do not assert uniform accuracy over the entire infinite time axis.

7 Possible Generalizations

The forcing term can be modified by changing its amplitude, rate parameter, or shape. For example, one may consider

$$N'(t) = G_0 \frac{\lambda^m t^{m-1} e^{-\lambda t}}{(m-1)!} - k_1 N(t) - \beta N(t)^p, \tag{16}$$

where $p > 1$.

For nonnegative solutions and nonnegative loss coefficients, the same basic upper bound

$$N(t) \le N_0 + A_m(t) \le N_0 + G_0$$

continues to hold, provided the generalized problem has the required existence and uniqueness properties. For real p that is not an integer, the expression N^p is naturally defined here on $N \ge 0$.

Another variation replaces the single Erlang-type input with a finite sum of such inputs:

$$F(t) = \sum_{r=1}^{q} G_r \frac{\lambda_r^{m_r} t^{m_r - 1} e^{-\lambda_r t}}{(m_r - 1)!},$$

where q and the m_r are positive integers, $G_r \geq 0$, and $\lambda_r > 0$.

In this setting,

$$\int_0^\infty F(t)\,dt = \sum_{r=1}^{q} G_r.$$

The boundedness argument then gives

$$0 \leq N(t) \leq N_0 + \sum_{r=1}^{q} G_r$$

for nonnegative loss coefficients and nonnegative initial data.

These extensions allow the model to represent several distinct transient inputs or different forms of nonlinear dissipation. Their practical usefulness depends on the system being modeled and on the interpretation of the parameters.

8 Conclusion

This paper examines a nonlinear differential equation with a factorially normalized, Erlang-type forcing term and linear and quadratic loss mechanisms. Under natural nonnegativity assumptions, the initial-value problem has a unique global solution. The solution remains nonnegative and satisfies the bound

$$0 \leq N(t) \leq N_0 + G_0.$$

The cumulative input has an explicit expression, and the model admits an explicit solution when quadratic loss is absent. In the general nonlinear case, an integral representation and a transformation to a second-order linear equation provide alternative ways to analyze the dynamics.

The long-time results show that the state tends to zero when either linear or quadratic dissipation is present. In the absence of both loss mechanisms, the state approaches the initial value plus the total input. The analytical bounds also provide consistency checks for numerical approximations.

Further work could investigate parameter estimation, sensitivity to the shape parameter m, improved numerical schemes, and applications to specific physical or biological systems. Such investigations would help determine when this model is appropriate and which features of its solutions are most relevant in practice.

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References

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